

Programming Exercises 2

1. Draw the hierarchy chart and then plan the logic for a program needed by the sales manager of The Henry Used Car Dealership. The program will determine the profit on any car sold. Input includes the sale price and actual purchase price for a car. The output is the profit, which is the sale price minus the purchase price.

Revise the profit-determining program so that it runs continuously for any number of cars. The detail loop executes continuously while the sale price is not 0.

2. Draw the hierarchy chart and then plan the logic for a program that calculates a person's body mass index (BMI). BMI is a statistical measure that compares a person's weight and height. The program prompts a user for and accepts the user's height in inches. The program accepts the user's weight in pounds and converts the user's height to meters and weight to kilograms. Then, it calculates BMI as weight in kilograms divided by height in meters squared, and displays the results. There are 2.54 centimeters in an inch, 100 centimeters in a meter, 453.59 grams in a pound, and 1,000 grams in a kilogram. Use *named constants* whenever you think they are appropriate.

Revise the BMI-determining program to execute continuously until the user enters 0 for the height in inches.

3. Draw the hierarchy chart and design the logic for a program for the River Falls Homes Construction Company. Design a program that prompts the user for a lot number in the River Falls subdivision and data about the home to be built there, including number of bedrooms, number of bathrooms, and the number of cars the garage holds. Output is the price of the home, which is a \$50,000 base price plus \$17,000 for each bedroom, \$12,500 for each bathroom, and \$6,000 for each car the garage holds. The program accepts lot numbers continuously until 0 is entered. Use named constants where appropriate.
4. Draw a structured flowchart or write pseudocode that describes the process of guessing a number between 1 and 100. After each guess, the player is told that the guess is too high or too low. The process continues until the player guesses the correct number. Pick a number and have a fellow student try to guess it by following your instructions.